Question 1

Prove that $x^* = (1, 1/2, -1)$ is optimal for the optimization problem,

$$\begin{array}{ll} \text{minimize} & (1/2)x^T P x + q^T x + r \\ \text{subject to} & -1 \le x_i \le 1, \quad i = 1, 2, 3 \end{array}$$

where,

$$P = \begin{bmatrix} 13 & 12 & -2 \\ 12 & 17 & 6 \\ -2 & 6 & 12 \end{bmatrix}, \quad q = \begin{bmatrix} -22.0 \\ -14.5 \\ 13.0 \end{bmatrix}, \quad r = 1.$$

Let $f(x) = \frac{1}{2}x^T P x + q^T x + r$, $X = \{x \in R^3 : -1 \le x_i \le 1\}$.

Since P is PSD. f is convex.

$\Rightarrow$ $x^* \in X$ is optimal iff $\nabla f(x^*)^T (y - x^*) \ge 0$, $\forall y \in X$.

$\Rightarrow$ $\nabla f(x) = Px + q$. with $x^* = [1, \frac{1}{2}, -1]$,

$Px^* = [21, 14.5, -11]^T$, $\nabla f(x^*) = Px^* + q = [-1, 0, 2]^T$

$\Rightarrow$ Hence for $\forall y \in X$. $\nabla f(x^*)^T (y - x^*) = -(y_1 - 1) + 0 \cdot (y_2 - \frac{1}{2}) + 2(y_3 + 1)$

$\Rightarrow$ $y_1 \le 1$, $-(y_1 - 1) \ge 0$; $y_3 \ge -1$, $2(y_3 + 1) \ge 0$.

$\Rightarrow$ the sum is non-negative for all $y \in X$.

$\Rightarrow$ by first order condition, x^* is optimal.

Question 2

Give an explicit solution of each of the following LPs.
(a) Minimizing a linear function over an affine set.

$$\begin{array}{ll} \text{minimize} & c^T x \\ \text{subject to} & Ax = b \end{array}$$

hint: Suppose

if the system $Ax=b$ is infeasible. no solution.
Assume $Ax=b$ is feasible and pick any feasible x_0
if there's $d \neq 0$ where $Ad=0$ & $c^T d < 0$.
$\Rightarrow x_0 + td$ feasible for all $t \geq 0$; $c^T(x_0 + td) \to -\infty$.
unbounded.
else: $c^T d = 0$ for all $d \in \ker A$. $\Rightarrow$ for any feasible x, $c^T x = v^T A x = v^T b$
$\Rightarrow$ the objective is constant on $\{x : Ax = b\}$; every feasible point is optimal., with optimal value beg $v^T b$.

(b) Minimizing a linear function over a halfspace.

$$\begin{array}{ll} \text{minimize} & c^T x \\ \text{subject to} & a^T x \leq b, \end{array}$$

where $a \neq 0$.

if c is not a scalar multiple of a. choose d with $a^T d = 0$ and $c^T d < 0$, then $x + td$ stays feasible and $c^T(x+td) \to -\infty$, unbounded
elif $c = \lambda a$ ① $\lambda > 0$, take d where $a^T d < 0$, $a^T(x+td) \to -\infty$,
$\Rightarrow c^T(x+td) = \lambda a^T(x+td) \to -\infty$, unbounded
② $\lambda = 0$, objective is constant $\Rightarrow$ if feasible, optimal

③ $\lambda<0$, $\min \lambda a^T x \Longleftrightarrow \max a^T x$, where $a^T x \le b$.

$\Rightarrow$ optimal attained on the boundary.

$\Rightarrow$ any x with $a^T x = b$ is optimal, with optimal value being λb.

(c) Minimizing a linear function over a rectangle.

$$\begin{array}{ll} \text{minimize} & c^T x \\ \text{subject to} & l \preceq x \preceq u \end{array} \tag{1}$$

where l and u satisfy $l \preceq u$.

an optimal solution is $x_i^* = \begin{cases} l_i, & c_i > 0 \\ u_i, & c_i < 0, \\ \text{any value in } [l_i, u_i]. \end{cases}$ $i = 1, \cdots, n$.